Customer Shopping Behavior Analysis

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Tools: Python | MYSQL (SQL) | Power BI

1. Project Overview

This report presents a structured SQL-based analysis of customer shopping behavior using transactional data from 3,900 purchases spanning multiple product categories, demographics, and sales channels. The primary goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior that can guide strategic business decisions.

2. Problem Statement

A leading retail company wants to better understand its customers' shopping behavior in order to improve sales, customer satisfaction, and long-term loyalty. The management team has observed changes in purchasing patterns across demographics, product categories, and sales channels (online vs. offline).

They are particularly interested in uncovering which factors — such as discounts, reviews, seasons, or payment preferences — drive consumer decisions and repeat purchases.

> **Core Business Question:** *"How can the company leverage consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?"*

3. Dataset Summary

The dataset contains 3,900 rows and 18 columns capturing a comprehensive view of retail transactions.

Attribute	Detail
Total Records	3,900 transactions
Total Columns	18 features
Customer Demographics	Age, Gender, Location, Subscription Status
Purchase Details	Item, Category, Amount (USD), Season, Size, Color
Behavior Features	Discount, Promo Code, Previous Purchases, Frequency, Review Rating, Shipping
Missing Data	37 null values in Review Rating (imputed using category median)

4. Data Preparation & Modeling (Python)

Prior to SQL analysis, the raw dataset was cleaned and transformed in Python using pandas:

- **Data Loading:** Imported the dataset using pandas and performed initial exploration with df.info() and .describe().

	Customer ID	Age	Purchase Amount (USD)	Review Rating	Previous Purchases
count	3900.000000	3900.000000	3900.000000	3863.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750065	25.351538
std	1125.977353	15.207589	23.685392	0.716983	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

- **Missing Value Imputation:** Filled 37 missing Review Rating values using the median rating per product category.

- **Column Standardization:** Renamed all columns to small_case for consistent SQL compatibility.
- **Feature Engineering:** Created age_group (by binning ages) and purchase_frequency_days columns.
- **Redundancy Check:** Confirmed that promo_code_used was redundant with discount_applied and dropped it.
- **Database Integration:** Connected Python to MySQL and loaded the cleaned DataFrame for SQL analysis.

5. Data Analysis using SQL

Ten structured SQL queries were executed in MySQL to answer key business questions. Results and insights are documented below.

Q1. Revenue by Gender

What is the total revenue generated by male vs. female customers?

gender	revenue
Male	157890
Female	75191

Q2. High-Spending Discount Users

Which customers used a discount but still spent more than the average purchase amount?

customer_id	discount_applied	gender	purchase_amount
2	Yes	Male	64
3	Yes	Male	73
4	Yes	Male	90
7	Yes	Male	85
9	Yes	Male	97
12	Yes	Male	68
13	Yes	Male	72
16	Yes	Male	81

customer_detail 2

Output

Action Output

#	Time	Action	Message
2	17:40:56	SELECT gender , SUM(purchase_amount) as revenue from customer_detail group by gender LIMIT 0, 1000	2 row(s) returned
3	17:42:33	SELECT customer_id, discount_applied, gender, purchase_amount FROM customer_detail WHERE discount ...	839 row(s) returned

Q3. Top 5 Products by Average Review Rating

Which are the top 5 products with the highest average review rating?

item_purchased	avg_rating	total_reviews
Gloves	3.86	140
Sandals	3.84	160
Boots	3.82	144
Hat	3.8	154
Handbag	3.78	153

Q4. Shipping Type Comparison

Compare the average purchase amounts between Standard and Express shipping.

shipping_type	avg_amount
Express	60.4752
Standard	58.4602

Q5. Subscribers vs. Non-Subscribers

Do subscribed customers spend more? Compare average spend and total revenue.

subscription_status	total_costomer	avg_spend	total_revenue
Yes	1053	59.4919	62645
No	2847	59.8651	170436

Q6. Discount-Dependent Products

Which 5 products have the highest percentage of purchases with discounts applied?

item_purchased	discount_rate
Hat	50.0000
Sneakers	49.6552
Coat	49.0683
Sweater	48.1707
Pants	47.3684

Q7. Customer Segmentation

Segment customers into New, Returning, and Loyal based on total previous purchases.

customer_segment	Number of customer
Loyal	3116
Returing	701
New	83

Q8. Top 3 Products per Category

What are the top 3 most purchased products within each product category?

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

Q9. Repeat Buyers & Subscriptions

Are customers who are repeat buyers (more than 5 previous purchases) more likely to subscribe?

subscription_status	repeat_buyers
Yes	958
No	2518

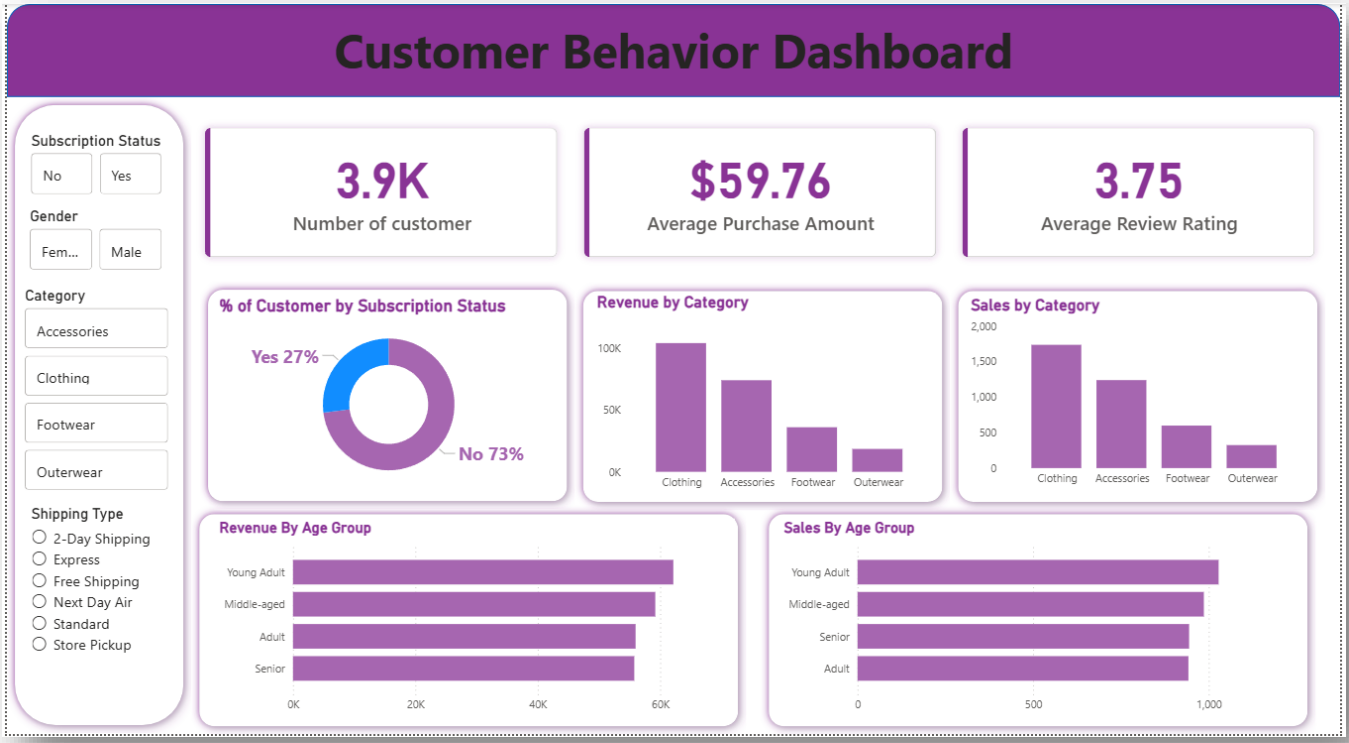
Q10. Revenue by Age Group

What is the revenue contribution of each age group?

age_group	total_revenue
Young Adult	62143
Middle-aged	59197
Adult	55978
Senior	55763

6. Dashboard in Power BI

Finally, we built an interactive dashboard in Power BI to present insights visually.



7. Business Recommendations

Based on the SQL analysis, the following strategic recommendations are proposed:

- **Boost Subscriptions:** Target repeat buyers (>5 purchases) with exclusive subscription benefits such as early access, free shipping, and loyalty points. This segment has the highest conversion potential.
- **Grow Female Customer Base:** Male customers currently generate over 2x the revenue of female customers. Develop female-focused campaigns, curated product lines, and influencer partnerships to close this gap.
- **Capitalize on Discount-Savvy Shoppers:** The 839 high-spending discount users represent a valuable segment. Offer personalized promotions or loyalty discounts to retain them rather than blanket discounting.
- **Revise Discount Policy for Key Products:** Hat, Sneakers, and Coat have ~50% discount rates. Shift from discounts to value-added offers (gift wrapping, bundles, warranties) to protect margins.
- **Promote Top-Rated Products:** Gloves, Sandals, and Boots lead in review ratings. Feature them prominently in homepage banners, email campaigns, and social ads to drive conversions.
- **New Customer Acquisition:** With only 83 New customers identified, launch targeted acquisition campaigns through referral programs, social media ads, and seasonal promotions to broaden the funnel.
- **Express Shipping Upsell:** Express shipping users spend more on average. Promote express shipping as a premium option at checkout with messaging around fast delivery and convenience.
- **Age-Targeted Marketing:** Young Adults lead in revenue but all age groups are closely matched. Personalize messaging per age cohort — digital campaigns for younger buyers, catalog/email for seniors.